

Quantum Approach to Solids - Problem Sheet 3

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* indicates harder question.

1. Heat capacity

- (a) Derive the Dulong-Petit law for the specific heat capacity of a solid, and state the limits of its validity.
- (b) To obtain a better model of the specific heat capacity, the atomic oscillations must be quantized. Compare and contrast the Einstein and Debye approaches to this problem.

2. Heat capacity in metals

At room temperature, do metals have a noticeably higher heat capacity than insulators because of the mobile electrons?

3. Einstein temperatures

Qualitatively, the melting point of a crystalline solid can be defined as the temperature T_m at which the amplitude of simple harmonic oscillations of the individual atoms about their lattice sites is about 10% of the interatomic spacing a (the Lindemann criterion). Show that in this (classical) picture, the frequency ν of the oscillations of a monatomic crystal (atomic weight M) satisfies the proportionality relation

$$\nu \propto \frac{1}{a} \left(\frac{T_m}{M} \right)^{1/2}.$$

The Einstein temperatures of aluminium and lead are 240 K and 67 K respectively. Guess orders of magnitude, or look up the data, to decide whether or not these temperatures and the above formula are in reasonable agreement.

4. Debye model (*)

Describe briefly the assumptions used in the Debye model for the vibrational energy of a solid and explain how they improve on the Dulong-Petit and Einstein models. What are the Debye frequency ω_D and the Debye temperature Θ_D .

Indicate how the model leads to a value of the internal energy U given by

$$U = \int_0^{\omega_D} \frac{3V\omega^2}{2\pi^2 c^3} \frac{\hbar\omega}{e^{\hbar\omega/k_B T} - 1} d\omega.$$

Define the various quantities in this equation.

Sketch the integrand as a function of $X = \hbar\omega/k_B T$ showing the behaviour at low and high frequencies. Show that the maximum contribution to the integral is $X \approx 3$.

Show that for a simple cubic crystal of atom spacing a , the maximum contribution to the integral comes from phonons of wavelength

$$\lambda_m = 0.5 \frac{\Theta_D}{T} a.$$

What does this suggest about the applicability of the Debye model?

[Note: the following expression may prove useful: $3N = \frac{V\omega_D^3}{2\pi^2 c^3}$.]

5. Thermal conductivity

Why does the size of a sample affect the thermal conductivity at low temperatures? Please define what is meant by low temperatures in your answer.

6. Heat transport along a wire

A well lagged wire of length L and cross-sectional area A has its ends maintained at temperatures T_1 and T_2 . The thermal conductivity of the wire is given by $\kappa = B + CT$, where T is the temperature and B and C are constants. What is the rate of heat flow along the wire?

7. Thermal conductivity in sapphire (*)

The thermal conductivity of an artificial sapphire crystal 3 mm in diameter reaches a sharp maximum at 30 K. Estimate roughly:

- (a) the maximum value of the thermal conductivity.
- (b) the thermal conductivity at liquid nitrogen temperatures (80 K). Explain the reasoning behind your estimates.

[For sapphire, $\Theta_D = 1000$ K, speed of sound $= 10^4$ ms⁻¹ and for $T \ll \Theta_D$, $C_v = 10^{-1} T^3$ Jm⁻³K⁻¹.]